<u>Problem</u>: calculate average particle density in the <u>ground</u> state of free Fermi gas (N fermions inside volume V).

$$\hat{\rho}(r) = \hat{\psi}^+(r)\hat{\psi}(r) = \frac{1}{V}\sum_{k_1 k_2} e^{i(\bar{k}_2-\bar{k}_1)\cdot\bar{r}}\, \hat{a}^+_{k_1}\hat{a}_{k_2}$$

Ground state of Fermi gas:

$$|\psi_0\rangle = \prod_{k} \hat{a}^+_k |0\rangle$$

↳ vacuum

$|k| \le k_F$ (i.e. product over the filled states)

Fermi momentum $k_F = p_F/\hbar$ (p_F = Fermi momentum)

Sum over filled states = total number of particles

$$\sum_{k\ (k\le k_F)} 1 = N \Rightarrow V\int \frac{d^3\vec{k}}{(2\pi)^3} = N$$

$$\frac{4\pi}{(2\pi)^3}\int_0^{k_F} dk\, k^2 = \frac{1}{2\pi^2}\frac{k_F^3}{3} = \frac{N}{V}$$

$$\Rightarrow k_F = \left(6\pi^2\frac{N}{V}\right)^{1/3}.$$

Average density

$$\bar{\rho} \equiv \langle\psi_0|\hat{\rho}(\bar{r})|\psi_0\rangle = \frac{1}{V}\sum_{k_1 k_2} e^{i(\vec{k}_2-\vec{k}_1)\cdot\vec{r}}$$

$$\cdot\langle\psi_0|\hat{a}^+_{k_1}\hat{a}_{k_2}|\psi_0\rangle$$

Now $\langle\psi_0|\hat{a}^+_{k_1}\hat{a}_{k_2}|1_1 1_2 1_3 \dots 1_{k_2-1} 1_{k_2} 1 \dots\rangle =$

$$= \langle\psi_0|\hat{a}^+_{k_1}(-1)^{\sum_{j=1}^{k_2-1} N_j}|1_1 1_2 \dots 1_{k_2-1} 0_{k_2} 1_{k_2+1}\rangle$$

But $\langle \Psi_0 | \hat{a}^+_{k_1} = \langle 1_1 1_2 1_3 \dots 1_{k_1-1} 1_{k_1} 1_{k_1+1} \dots | \hat{a}^+_{k_1} =$

$$= \langle 1_1 1_2 \dots 1_{k_1-1} 0_{k_1} 1_{k_1+1} \dots | (-1)^{\sum_{e=1}^{k_1-1} N_e}$$

(Because $\langle \Psi_0 | \hat{a}^+_{k_1} = \left(\hat{a}_{k_1} | \Psi_0 \rangle\right)^+$)

Hence $\langle \Psi_0 | \hat{a}^+_{k_1} \hat{a}_{k_2} | \Psi_0 \rangle = \langle 1_1 1_2 \dots 1_{k_1-1} 0_{k_1} 1 \dots |$

$$(-1)^{\sum N_e} (-1)^{\sum N_j} | 1_1 1_2 \dots 1_{k_2-1} 0_{k_2} 1 \dots \rangle =$$

$$= \left((-1)^{\sum N_j}\right)^2 \langle 1_1 1_2 \dots 0_{k_1} \dots | 1_1 1_2 \dots 0_{k_1} \rangle = 1$$

for $k_1 = k_2$ and also $k_1 \leq k_F$.

and $= 0$ for $k_1 \neq k_2$.

$$\Rightarrow \bar{\rho} = \frac{1}{V} \sum_{\substack{k_1 k_2 \\ k_1 \leq k_F}} e^{i(\vec{k}_2 - \vec{k}_1) \cdot \vec{r}} \delta_{k_1 k_2} \cdot 1$$

$$= \frac{1}{V} \sum_{k_1 \leq k_F} 1 = \frac{N}{V} \text{ by def.}$$

The lesson:

$$\langle \Psi_0 | \hat{a}^+_{k_1} \hat{a}_{k_2} | \Psi_0 \rangle = \begin{cases} \delta_{k_1, k_2} & \text{for } k_1 \leq k_F \\ 0 & \text{for } k_1 > k_F \end{cases} =$$

$$= \delta_{k_1, k_2} \cdot (\text{occupation number of state } k_1)$$

Now recall that electrons have spin, $\hat{a}_k \to \hat{a}_{k,\sigma}$

where $|\sigma\rangle$ is : $\hat{S}_z |\sigma\rangle = \frac{\hbar}{2}\sigma|\sigma\rangle$, $\sigma = \pm 1$

eigenstate of spin operator

$$\hat{\Psi}(r) \to \hat{\Psi}_\sigma(r) = \sum_k \hat{a}_k \varphi_k(\vec{r}) \chi_\sigma$$

where χ_σ is the spinor, e.g. $\chi_{+1} = \begin{pmatrix}1\\0\end{pmatrix}$, $\chi_{-1} = \begin{pmatrix}0\\1\end{pmatrix}$ etc...

$$\hat{\rho}(\vec{r}) = \sum_\sigma \hat{\rho}(\vec{r},\sigma) = \frac{1}{V}\sum_\sigma \sum_{k_1,k_2} e^{i(\vec{k}_2-\vec{k}_1)\cdot\vec{r}}\, \hat{a}^+_{k_1\sigma}\hat{a}_{k_2\sigma}$$

and $|\Psi_0\rangle = \prod_{\substack{k\le k_F\\ \sigma=\pm1}} \hat{a}^+_{k,\sigma}|0_\uparrow, 0_\downarrow\rangle$ i.e. two electrons with opposite spins ($\sigma = \pm 1$) for each occupied $\vec{k}$ state.

k_F found from: $\sum_{\substack{k\le k_F\\ \sigma}} 1 = N \Rightarrow \frac{(2s+1)k_F^3}{6\pi^2} = \frac{N}{V}$

$$k_F = \left(\frac{6\pi^2}{2s+1}\frac{N}{V}\right)^{1/3}$$

For electrons $s = \frac{1}{2}$ and $2s+1 = 2$ of course.

$$\bar{\rho} = \langle\Psi_0|\hat{\rho}(\vec{r})|\Psi_0\rangle = \frac{1}{V}\sum_{\substack{k\le k_F\\ \sigma}} 1 = \frac{N}{V}$$ (of course the same as for $s=0$)

Of course

$$\bar{\rho}_\sigma = \langle\Psi_0|\hat{\rho}(\vec{r},\sigma)|\Psi_0\rangle = \frac{N}{(2s+1)V} = \frac{N}{2V}.$$

no summation over $\underline{\sigma}$ here

$\bar{\rho}_\sigma$ = density per spin component

Problem

Find the ground state correlation function of densities of ~~particle~~ fermions with spin projection σ_1, $\hat{\rho}(\vec{r}_1, \sigma_1)$, with that at point $\vec{r}_2$ and spin projection σ_2, $\hat{\rho}(\vec{r}_2, \sigma_2)$

$$\hat{\rho}(\vec{r}_1, \sigma_1) = \frac{1}{V} \sum_{k_1 k_2} e^{i(\vec{k}_2 - \vec{k}_1)\vec{r}_1} \hat{a}^+_{k_1 \sigma_1} \hat{a}_{k_2 \sigma_1}$$

$$\hat{\rho}(\vec{r}_1, \sigma_1)\, \hat{\rho}(\vec{r}_2, \sigma_2) = \frac{1}{V^2} \sum_{\substack{k_1 k_2 \\ k_3 k_4}} e^{i\vec{r}_1(\vec{k}_2 - \vec{k}_1) + i\vec{r}_2(\vec{k}_4 - \vec{k}_3)}$$

$$\hat{a}^+_{k_1 \sigma_1} \hat{a}_{k_2 \sigma_1} \hat{a}^+_{k_3 \sigma_2} \hat{a}_{k_4 \sigma_2}$$

Need $\langle \Psi_0 | \hat{\rho}(r_1, \sigma_1) \hat{\rho}(r_2, \sigma_2) | \Psi_0 \rangle$.

1) let $\sigma_2 \neq \sigma_1$. Then since $|\Psi_0\rangle = \prod_{\substack{k \leq k_F \\ \sigma}} \hat{a}^+_{k\sigma} |0\rangle =$

$$= \left(\prod_{\substack{k \leq k_F \\ \sigma = +1}} \hat{a}^+_{k, \sigma=+1} |0\rangle_{\sigma=+1} \right) \left(\prod_{\substack{k \leq k_F \\ \sigma = -1}} \hat{a}^+_{k, \sigma=-1} |0\rangle_{\sigma=-1} \right) =$$

$= |\Psi_0\rangle_{\sigma=+1} |\Psi_0\rangle_{\sigma=-1}$, the correlation function reduces to the product

$$\langle \Psi_{0, \sigma=\sigma_1} | \hat{\rho}(r_1, \sigma_1) | \Psi_{0, \sigma=\sigma_1} \rangle \cdot \langle \Psi_{0, \sigma=\sigma_2} | \hat{\rho}(r_2, \sigma_2) | \Psi_{0, \sigma_2} \rangle$$

$$= \left(\frac{N}{2V} \right)^2.$$

Physical reason: fermions with $\sigma_1 \neq \sigma_2$ behave as distinguishable particles

2) Now $\sigma_1 = \sigma_2 = \sigma$ (and we can supress spin index altogether)

$$\frac{1}{V^2} \sum_{(k_i)} e^{i\vec{r}_1(\vec{k}_2-\vec{k}_1)+i\vec{r}_2(\vec{k}_4-\vec{k}_3)} \langle \Psi_0 | \hat{a}^+_{k_1\sigma} \hat{a}_{k_2\sigma} \hat{a}^+_{k_3\sigma} \hat{a}_{k_4\sigma} | \Psi_0 \rangle$$

$$\hat{a}^+_{k_1} \underbrace{\hat{a}_{k_2} \hat{a}^+_{k_3}}_{\delta_{k_2k_3} - \hat{a}^+_{k_3}\hat{a}_{k_2}} \hat{a}_{k_4} = \underbrace{\delta_{k_2k_3} \hat{a}^+_{k_1} \hat{a}_{k_4}}_{\text{"A"}} - \underbrace{\hat{a}^+_{k_1} \hat{a}^+_{k_3} \hat{a}_{k_2} \hat{a}_{k_4}}_{\text{"B"}}$$

But $\langle \Psi_0 | \hat{a}^+_{k_1} \hat{a}_{k_4} | \Psi_0 \rangle = \delta_{k_1,k_4} \begin{cases} \text{for } k_1 \le k_F \\ 0, \text{ otherwise} \end{cases}$

$\Rightarrow$ the first term ("A") just gives

$$\frac{1}{V^2} \sum_{\substack{k_2=k_3 \\ k_1=k_4}} e^{i\vec{r}_1(\vec{k}_2-\vec{k}_1)+i\vec{r}_2(\vec{k}_1-\vec{k}_2)}$$

Two possibilities only:

1) $k_4 = k_3$ and $k_4 \le k_F$; $k_2 = k_1$ and $k_2 \le k_F$

2) $k_4 = k_1$ and $k_4 \le k_F$; $k_3 = k_2$ and $k_3 \ge k_F$ (!)

A $\rightarrow$ $$\frac{1}{V^2} \sum_{\substack{k_1,k_3 \\ k_{1,3} \le k_F}} \langle \Psi_0 | \hat{a}^+_{k_1\sigma} \hat{a}_{k_1\sigma} \hat{a}^+_{k_3\sigma} \hat{a}_{k_3\sigma} | \Psi_0 \rangle$$

B $\rightarrow$ $$+\frac{1}{V^2} \sum_{\substack{k_1,k_3 \\ (k_1 \le k_4 \text{ but } k_3 \ge k_F)}} e^{i\vec{r}_1(\vec{k}_3-\vec{k}_1)+i\vec{r}_2(\vec{k}_1-\vec{k}_3)} \langle \Psi_0 | \hat{a}^+_{k_1\sigma} \hat{a}_{k_3\sigma} \hat{a}^+_{k_3\sigma} \hat{a}_{k_1\sigma} | \Psi_0 \rangle$$

First term (A) $= \left(\frac{1}{V}\sum_{k_1\le k_F} \langle\Psi_0|\hat{a}^+_{k_1\sigma}\hat{a}_{k_1\sigma}|\Psi_0\rangle\right)\cdot$

$$\cdot\left(\frac{1}{V}\sum_{k_3\le k_F}\langle\Psi_0|\hat{a}^+_{k_3\sigma}\hat{a}_{k_3\sigma}|\Psi_0\rangle\right) +$$

$$+\frac{1}{V^2}\sum_{\substack{k_1=k_3\\ k_1\le k_F}}\langle\Psi_0|\hat{a}^+_{k_1}(1-\hat{a}^+_{k_1}\hat{a}_{k_1})\hat{a}_{k_1}|\Psi_0\rangle =$$

but $\hat{a}_{k_1}\hat{a}_{k_1}|\Psi_0\rangle = 0$

$$= \frac{N/2}{V}\cdot\frac{N/2}{V} + \frac{N/2}{V^2} = \left(\frac{N}{2V}\right)^2 + \frac{N}{2V}\cdot\frac{1}{V} \xrightarrow[\substack{V\to\infty\\ \text{but } \frac{N}{V}=\text{const.}}]{N\to\infty} \left(\frac{N}{2V}\right)^2$$

Second term (B) $= \frac{1}{V^2}\sum_{\substack{k_1\le k_F\\ k_3\ge k_F}} e^{i(\vec{r}_1-\vec{r}_2)(\vec{k}_3-\vec{k}_1)}\cdot$

$$\cdot\langle\Psi_0|\hat{a}^+_{k_1\sigma}\hat{a}_{k_1\sigma}\hat{a}_{k_3\sigma}\hat{a}^+_{k_3\sigma}|\Psi_0\rangle \quad (=1)$$

Let $\vec{r} = \vec{r}_1 - \vec{r}_2$.

Note that $\frac{1}{V}\sum_{k_3\ge k_F} e^{i\vec{r}\cdot\vec{k}_3} = \frac{1}{V}\sum_{k_3} e^{i\vec{r}\cdot\vec{k}_3} -$

(no restrictions on $\vec{k}_3$ here)

$$-\frac{1}{V}\sum_{k_3\le k_F} e^{i\vec{r}\cdot\vec{k}_3} = \delta(\vec{r}) - \frac{1}{V}\sum_{k_3\le k_F} e^{i\vec{r}\cdot\vec{k}_3}$$

because $\frac{1}{V}\sum_{k_3} e^{i\vec{k}_3\vec{r}} = \int\frac{d\vec{k}_3}{(2\pi)^3}e^{i\vec{k}_3\cdot\vec{r}} = \delta(\vec{r})$.

Thus $B = \frac{1}{V} \sum_{k_1 \le k_F} e^{-i\vec{r}\cdot\vec{k}_1} \left\{ \delta(\vec{r}) - \frac{1}{V} \sum_{k_3 \le k_F} e^{i\vec{r}\cdot\vec{k}_3} \right\} =$

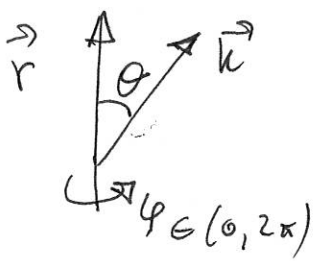
$$= \delta(\vec{r}) \cdot \frac{1}{V} \sum_{k_1 \le k_F} 1 - \left(\frac{1}{V} \sum_{k_1 \le k_F} e^{-i\vec{r}\cdot\vec{k}_1} \right) \left(\frac{1}{V} \sum_{k_3 \le k_F} e^{i\vec{r}\vec{k}_3} \right)$$

equal!
(up to conjugation)

$$= \delta(\vec{r}_1 - \vec{r}_2) \cdot \frac{N/2}{V} - \left| \frac{1}{V} \sum_{k_1 \le k_F} e^{i\vec{r}\cdot\vec{k}_1} \right|^2 .$$

$$I = \frac{1}{V} \sum_{k \le k_F} e^{i\vec{k}\cdot\vec{r}} = \int_{k \le k_F} \frac{d\vec{k}}{(2\pi)^3} e^{i\vec{k}\cdot\vec{r}} = \int_0^{k_F} \frac{dk \cdot k^2}{(2\pi)^3} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \cdot$$

$$\cdot e^{ikr\cos\theta}$$



$$\int_0^{\pi} d\theta \sin\theta \, e^{ikr\cos\theta} = -\int_1^{-1} d(\cos\theta) \, e^{ikr\cos\theta} = \int_{-1}^{1} du \, e^{ikru} =$$

$$= \left. \frac{e^{ikru}}{ikr} \right|_{-1}^{1} = \frac{2i\sin(kr)}{ikr}$$

Thus $I = \frac{2}{(2\pi)^2} \int_0^{k_F} dk \, k^2 \frac{\sin(kr)}{kr} = \frac{1}{2\pi^2 r^3} \int_0^{k_F r} dx \, x^2 \frac{\sin(x)}{x}$

$x = kr$

$$\int_0^{k_F r} dx\, x \sin x = \int_0^{k_F r} d(\cos x)\, x = x\cos x \Big|_0^{k_F r} - \int_0^{k_F r} \cos x\, dx =$$

$$= k_F r \cos(k_F r) - \sin(k_F r)$$

$$I = \frac{1}{2\pi^2 r^3}\Big(k_F r \cos(k_F r) - \sin(k_F r)\Big)$$

Thus $\langle \Psi_0 | \hat{\rho}(r_1,\sigma)\hat{\rho}(r_2,\sigma) | \Psi_0 \rangle = \left(\frac{N/2}{V}\right)^2 + \delta(\vec{r}_1 - \vec{r}_2)\frac{N/2}{V}$

$$- \frac{1}{4\pi^4}\frac{\big(k_F r\cos(k_F r) - \sin(k_F r)\big)^2}{r^6}.$$

So that correlation function

$$\nu(\vec{r} = \vec{r}_1 - \vec{r}_2) = \langle \Psi_0 | \hat{\rho}(r_1,\sigma)\hat{\rho}(r_2,\sigma) | \Psi_0 \rangle - \left| \langle \Psi_0 | \hat{\rho}(r_1,\sigma) | \Psi_0 \rangle \right|^2$$

$$= \delta(\vec{r}_1 - \vec{r}_2)\frac{N}{2V} - \frac{\big(k_F r\cos(k_F r) - \sin(k_F r)\big)^2}{4\pi^4 r^6}.$$

Long-distance limit: $k_F r \gg 1 \Rightarrow \nu(r) \approx -\frac{k_F^2}{4\pi^4}\frac{\cos^2(k_F r)}{r^4}$

Short-distance limit: $k_F r \ll 1$

$$x\cos x - \sin x \underset{x\to 0}{\approx} x\left(1 - \frac{x^2}{2}\right) - \left(x - \frac{x^3}{6}\right) = \left(\frac{1}{6} - \frac{1}{2}\right)x^3 = -\frac{x^3}{3}$$

$$\Rightarrow \nu(r) \underset{r\to 0}{\approx} -\frac{1}{4\pi^4 r^6}\left(-\frac{(k_F r)^3}{3}\right)^2 = -\frac{k_F^6}{\pi^4}\left(\frac{1}{4}\cdot\frac{1}{9}\right) \text{ (constant value)}$$

But $k_F^3 = 3\pi^2 \frac{N}{V} \Rightarrow k_F^6 = 9\pi^4 \left(\frac{N}{V}\right)^2$

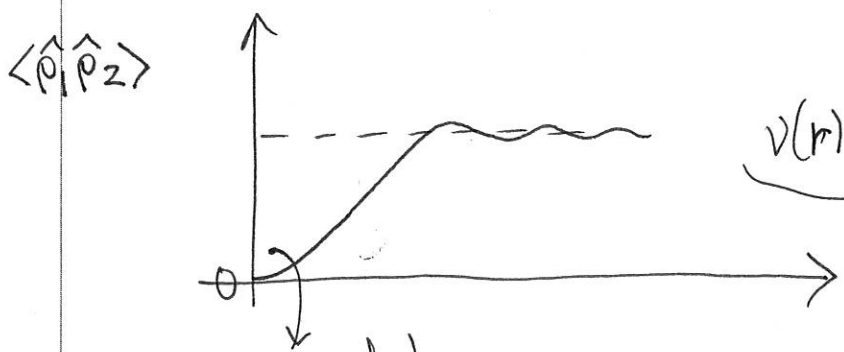
$$\frac{k_F^6}{4 \cdot 9\pi^4} = \left(\frac{N/2}{V}\right)^2$$

So that $\langle \Psi_0 | \hat{\rho}(\vec{r}_1, \sigma)\, \hat{\rho}(\vec{r}_1 + \vec{r}, \sigma) | \Psi_0 \rangle \xrightarrow[r \to 0]{} \left(\frac{N/2}{V}\right)^2 + \delta(\vec{r}) \frac{N}{2V}$

$\vec{r}_2 = \vec{r}_1 + \vec{r}$

$$- \left(\frac{N/2}{V}\right)^2 = \delta(\vec{r}) \frac{N}{2V} = 0 \text{ for } r \to 0$$

but away from exact $\vec{r} = 0$ point.



correlation hole for fermions of the same spin!

$$v(r) = -\frac{k_F^6}{4\pi^4} \frac{\cos^2(k_F r)}{(k_F r)^6} \text{ for } k_F r \gg 1$$

$$= -\frac{9}{2}\left(\frac{N}{2V}\right)^2 \frac{1 + \cos(2k_F r)}{(k_F r)^6}$$

Note oscillations. $\cos(2k_F r)$!

Note the appearance of $\cos(2k_F r)$ term: Friedel oscillations of electron density due to the existence of Fermi surface!

Fermi gas in a box $\longrightarrow$ open boundary conditions

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\varphi = \varepsilon\varphi \qquad \varphi(0)=\varphi(L)=0$$

$$\varphi(x) = A\sin(kx)$$

$$\sin(kL)=0 \Rightarrow k_n = \frac{\pi n}{L}, \quad \underline{n>0} \quad n\in\mathbb{Z}$$

Norm. $$A^2\int_0^L dx\,\sin^2(kx) = 1$$

$$\frac{A^2}{2}\int_0^L dx\,(1-\cos 2kx) = \frac{A^2}{2}\left(L - \frac{\sin 2kx}{2k}\Big|_0^L\right) = \frac{L}{2}A^2$$

$$A = \sqrt{\frac{2}{L}}$$

$$\varphi_n(x) = \sqrt{\frac{2}{L}}\,\sin\left(\frac{\pi n x}{L}\right)$$

$$\hat{\psi}(x) = \sum_n \varphi_n(x)\,\hat{a}_n$$

$$\hat{\rho}(x) = \hat{\psi}^+(x)\,\hat{\psi}(x) = \frac{2}{L}\sum_{n,n'}\sin\left(\frac{\pi n x}{L}\right)\sin\left(\frac{\pi n' x}{L}\right)\hat{a}_n^+\hat{a}_{n'}$$

$$\langle\psi_0|\hat{\rho}(x)|\psi_0\rangle = \frac{2}{L}\sum_n \sin^2\left(\frac{\pi n x}{L}\right)\langle\psi_0|\hat{a}_n^+\hat{a}_n|\psi_0\rangle$$

$$|\psi_0\rangle = \prod_{|m|\le n_F}\hat{a}_m^+\,|0\rangle$$

where $$\sum_{n=1}^{n_F} 1 = N \Rightarrow n_F = N.$$

$$\sum_n 1 = \int_{-k_F}^{k_F} \frac{dk}{\Delta k} = \frac{L}{\pi}\int_0^{k_F} dk = \frac{L}{\pi} k_F = N \Rightarrow k_F = \frac{\pi N}{L}$$

with $\Delta k = \frac{\pi}{L}$

$$\bar{\rho}(x) = \frac{2}{L}\sum_{n \le n_F} \sin^2\left(\frac{\pi n x}{L}\right) = \frac{2}{L}\frac{L}{\pi}\int_{-k_F}^{k_F} dk\, \sin^2(kx) =$$

$$= \frac{2}{\pi}\int_{-k_F}^{k_F} dk\, \frac{1}{2}(1-\cos 2kx) = \frac{1}{\pi}\int_0^{k_F} dk\,(1-\cos 2kx) =$$

$$= \frac{1}{\pi}\left(k_F - \frac{\sin(2kx)}{2x}\Big|_0^{k_F}\right) = \frac{1}{\pi}\left(k_F - \frac{\sin(2k_F x)}{2x}\right)$$

$$= \frac{1}{\pi}k_F\left(1 - \frac{\sin(2k_F x)}{2k_F x}\right) = \rho_0\left(1 - \frac{\sin(2k_F x)}{2k_F x}\right)$$

but $k_F = \frac{\pi}{2}\frac{N}{L} \equiv \frac{\pi}{2}\rho_0$

Friedel oscillations with spatial period $\frac{\pi}{k_F}$.

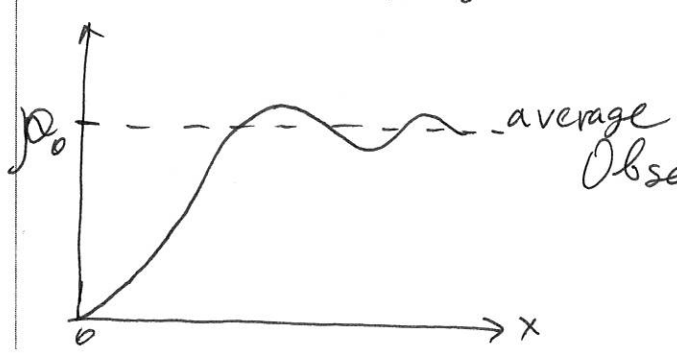
Note the limits:

$$\bar{\rho}(x\to 0) \Rightarrow \rho_0\left(1 - \frac{2k_F x}{2k_F x}\right) = 0.$$

~~$\bar{\rho}(x\to L) = \rho_0\, 1 - \sin$~~ $\bar{\rho}(x\to L) = 0$ too

$x = L - y$

but to see this one needs to sum over (n) more accurately!



Observe non-trivial density profile caused by hard wall boundary condition.

$$\bar{\rho}(x) = \frac{2}{L}\sum_{n=-n_F}^{n_F} \sin^2\left(\frac{\pi n x}{L}\right) = \frac{2}{L}\left\{ \cancel{\sum_{n=-n_F}^{0} \sin^2\left(\frac{\pi n x}{L}\right)} + \right.$$

$$\left. + \sum_{0}^{n_F} \sin^2\left(\frac{\pi n x}{L}\right) - \cancel{\sin^2\left(\frac{\pi n x}{L}\right)}\Big|_{n=0} \right\} = \frac{2}{L}\sum_{n=0}^{n_F} \sin^2\left(\frac{\pi n x}{L}\right)$$

$$n_F = \frac{L}{\pi} k_F = N.$$

$$= \frac{1}{L}\sum_{n=0}^{n_F}\left\{1 - \cos\left(\frac{2\pi n}{L}x\right)\right\}$$

$$\sum_{n=0}^{n_F} e^{i\frac{2\pi x}{L}n} = \underset{0}{1} + \underset{1}{e^{i\frac{2\pi x}{L}}} + \underset{2}{e^{i\frac{4\pi x}{L}}} + \dots =$$

$$= \sum_{0}^{n_F}\left(e^{i\frac{2\pi x}{L}}\right)^{n} = \frac{1-\left(e^{i\frac{2\pi x}{L}}\right)^{n_F+1}}{1-e^{i\frac{2\pi x}{L}}}$$

$$\bar{\rho}(x) = \frac{1}{L}\left(n_F+1 - \mathrm{Re}\left\{\frac{1-\left(e^{i\frac{2\pi x}{L}}\right)^{n_F+1}}{1-e^{i\frac{2\pi x}{L}}}\right\}\right)$$

$$\frac{e^{i\frac{\pi x}{L}(n_F+1)}\left(e^{-i\frac{\pi x}{L}(n_F+1)} - e^{i\frac{\pi x}{L}(n_F+1)}\right)}{e^{i\frac{\pi x}{L}}\left(e^{-i\frac{\pi x}{L}} - e^{i\frac{\pi x}{L}}\right)} =$$

$$= e^{i\frac{\pi x}{L}n_F}\,\frac{\sin\frac{\pi x}{L}(n_F+1)}{\sin\frac{\pi x}{L}}$$

$$\cancel{n_F = \frac{N}{2}}$$

$$\cancel{n_F + 1 = \frac{N+1}{2}}$$

Obtain n_F: $\cancel{\sum_{-n_F}^{n_F} 1 = 2\sum_{0}^{n_F} 1 - 1 = 2(n_F+1) - 1 = 2n_F + 1 = N}$

$\cancel{n_F = \frac{N-1}{2}}$

$n_F = N$

$\cancel{n_F = \frac{N+1}{2}}$

Now expectation value

$$\bar{\rho}(x) = \frac{N+1}{L}\left\{1 - \frac{1}{N+1}\,\frac{\cos\left(\frac{\pi x}{L}\cdot N\right)\sin\left(\frac{\pi x}{L}\cdot\frac{N+1}{}\right)}{\sin\left(\frac{\pi x}{L}\right)}\right\}$$

$\cancel{N = \text{odd so let } N = 2k+1, \quad \frac{N-1}{2} = k}$

$\cancel{\uparrow \text{ integer} \qquad \frac{N+1}{2} = k+1}$

$$\bar{\rho}(x) = \frac{N+1}{L}\left\{1 - \frac{1}{N+1}\,\frac{\cos\left(\frac{\pi N}{L}x\right)\sin\left(\frac{\pi(N+1)}{L}x\right)}{\sin\left(\frac{\pi}{L}x\right)}\right\}$$

let $x = L - y \Rightarrow \cos\frac{\pi N}{L}(L-y) = \cos\left(\pi N - \frac{\pi N y}{L}\right) \to (-1)^N \cos\frac{\pi N y}{L}$

$$\sin\frac{\pi(N+1)}{L}(L-y) = \sin\left(\pi(N+1) - \frac{\pi(N+1)y}{L}\right) =$$

$$= -\cos\pi(N+1)\,\sin\frac{\pi(N+1)y}{L} = -(-1)^{N+1} \cdot \sin\frac{\pi(N+1)y}{L}$$

$$\sin\frac{\pi}{L}(L-y) = \sin\left(\pi - \frac{\pi y}{L}\right) = \sin\frac{\pi y}{L}$$

$$\Rightarrow 1 - \frac{1}{N+1}\,\frac{(-1)^N(-1)^{N+2}\sin\left(\frac{\pi(N+1)y}{L}\right)}{\sin\frac{\pi y}{L}} = 1 - \frac{\frac{\pi(N+1)y}{L}}{\frac{\pi y}{L}}\,\frac{1}{N+1} = 0$$